

RETHINKING MOTION – FROM SPIN TO INTELLIGENCE

A Power-Upgraded, Real-Time Sensor-Based Robotic Spinning-Top Platform

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Small Body. Real-Time Mind.

What if a spinning top could become aware of its own dynamics – sensing, logging, streaming and displaying its behavior in real-time?

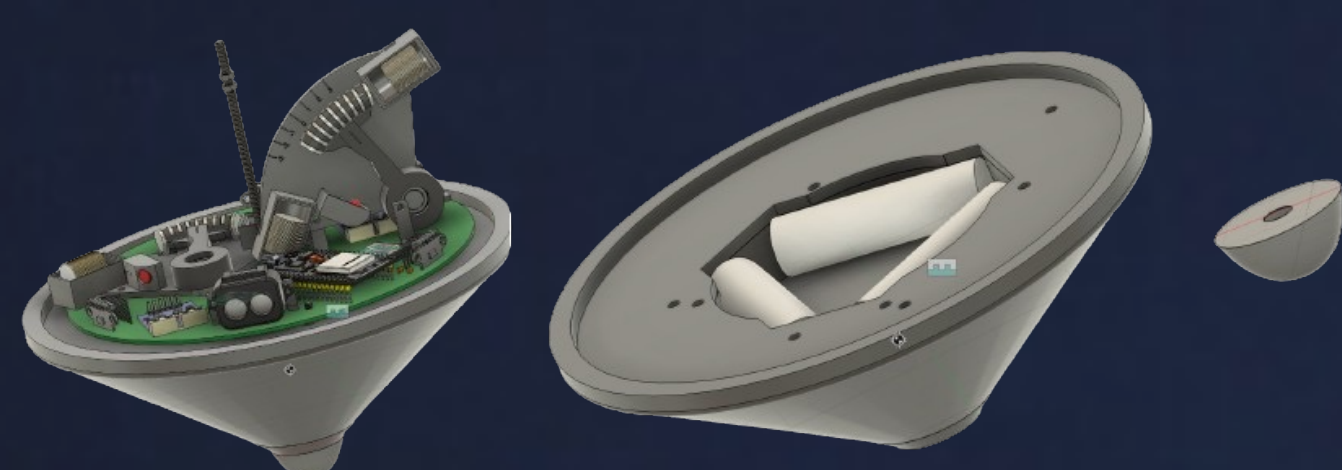
Spin. Sense. Sync. The next generation of motion starts here.

VISION

A compact rotating body – evolving toward autonomous tactile robotics.

A step toward a fully autonomous spinning robot capable of maintaining balance, reacting to its environment, communicating its state, and coordinating motion through real-time feedback control.

3D MODEL



THE BIG IDEA

Project A built the physical platform.

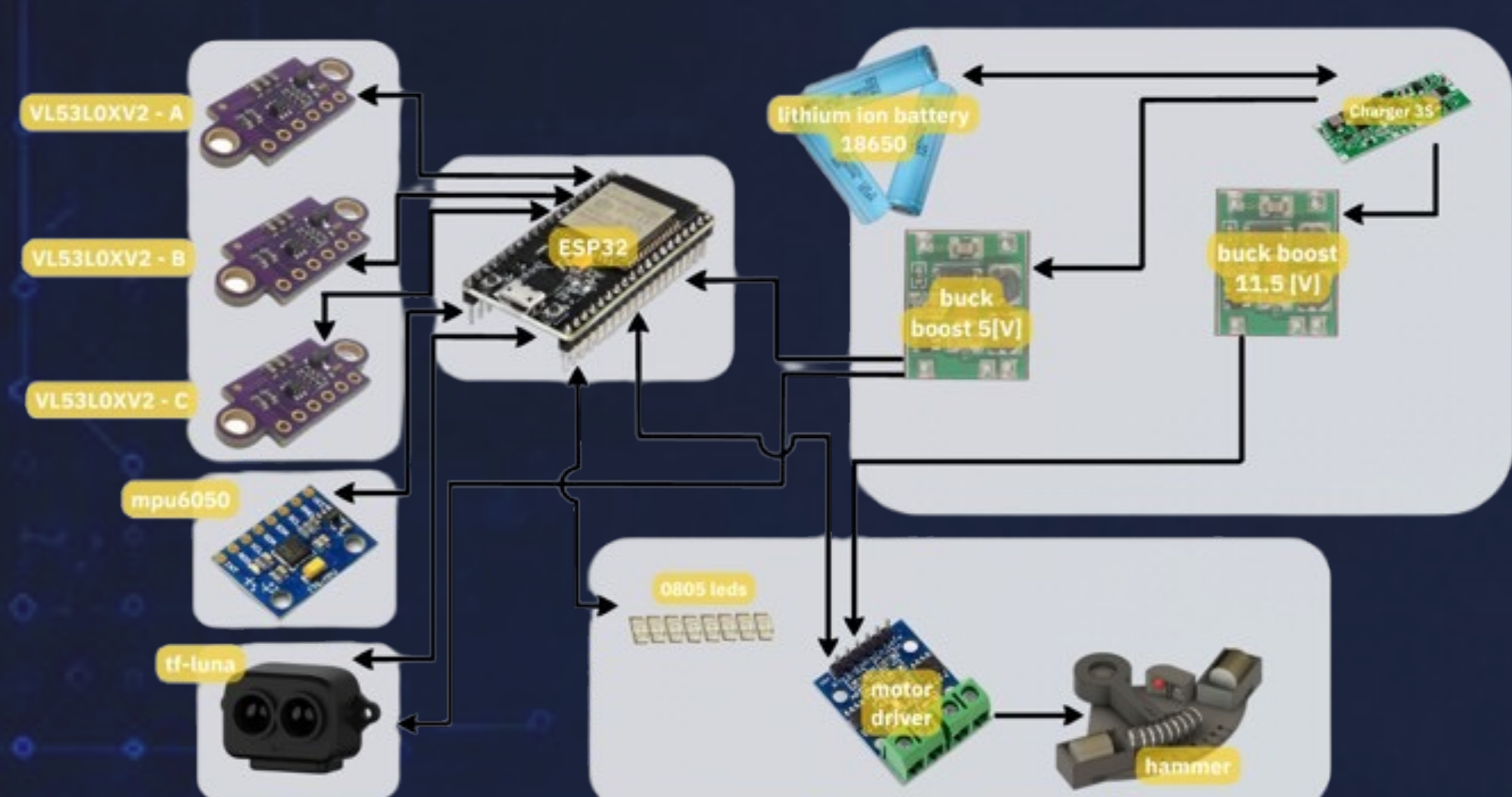
Project B made it powerful, measurable, synchronized and ready for control

- Real-Time Rotating Embedded System
- Multi-Sensor Spatial and Motion Awareness
- High-Power Hammer Actuation
- Wireless Telemetry and Android Visualization
- POV Display Synchronized with Rotation
- Experimental Platform for Future Active Stabilization

CORE IMPROVEMENTS

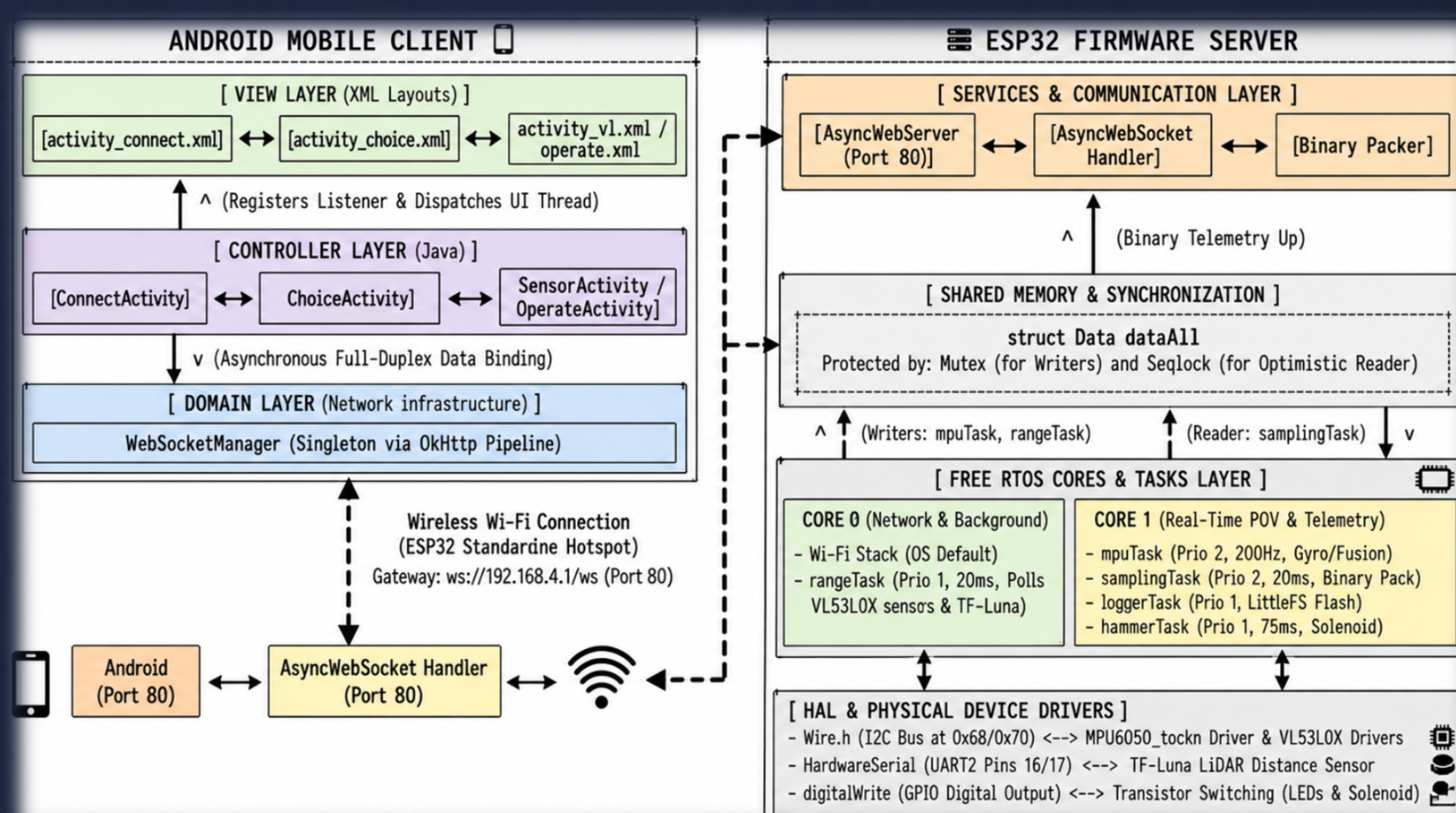
- 3S Power Transformation: upgraded from 1S prototype to 3-cell power architecture for stronger actuation
- Real-Time Firmware Architecture: implemented concurrent ESP32 task execution for sensor acquisition, logging, telemetry and POV timing
- Live Telemetry Platform: streamed synchronized system data to an Android application using WebSocket communication
- Experimental Data Pipeline: logged high-density CSV measurements for offline debugging, validation, and performance analysis
- Mechanical Rebalancing: redesigned the internal structure and balanced the rotating body after the power upgrade
- POV Synchronization: developed angle-based LED timing for visual output during rotation
- Engineering Failure Analysis: identified hardware bottlenecks including GPIO conflicts, power instability, and timing constraints

HARDWARE ARCHITECTURE



Architecture Flow:
Sensors->Real-Time ESP32 Processing->Control/Logging/ Telemetry/ POV display

SOFTWARE ARCHITECTURE



- Interfaces:
WebSocket,
HTTP/CSV,
Android application,
LED POV display

POV DISPLAY – VISUALIZING ROTATION

The spinning-motion itself becomes the display surface

- Angle-synchronized LED activation
- Digit display based on spin count
- Short LED pulses timed to rotational position
- Sensitive to delays, blocking code, and communication stalls
- Demonstrates the connection between motion, timing, and visual perception

REAL-TIME SOFTWARE ENGINE

The nervous system of the spinning-top:

Sensing motion, timing visual output, logging experiments, and communicating with the outside world in real time

- FreeRTOS-based concurrent task execution
- High-frequency gyroscope sampling and angle tracking
- Non-blocking telemetry and logging design
- Protected shared variables for task synchronization
- Reduced timing jitter in POV display logic
- Separation between control-critical and background operations

LIVE TELEMTRY & ANDROID VISUALIZATION

Real-Time telemetry turns invisible dynamics into visible system behavior

- High-density CSV logging for accurate offline analysis
- HTTP access for retrieving logged experiment file
- Throttled WebSocket streaming for smooth real-time visualization
- Live graphs for monitoring sensor and system behavior
- Bidirectional control for modes and hammer commands

INTEGRATED SYSTEM CHALLENGES AND OPTIMIZATION

Power, timing, telemetry, and mechanical stability – optimized as one rotating embedded system

Hardware & Mechanical Lessons

- GPIO strapping/debug pins limited reliable motor-driver control
- High-current actuation introduced power and EMI sensitivity
- Mechanical asymmetry required iterative physical balancing
- Future scalability requires safer pin mapping and I/O expansion

Software & Real-Time Lessons

- High-frequency execution had to coexist with high-throughput telemetry and logging
- 200 Hz MPU sampling, 50 Hz distance polling, WebSocket streaming, and CSV logging created multi-rate timing constraints
- The 1 ms POV rendering loop required non-blocking access to synchronized data snapshots
- Binary telemetry replaced high-frequency JSON to reduce bandwidth, latency, and Android overload
- Ping-pong buffering preserved full-resolution CSV logging without blocking real-time behavior
- Telemetry throttling and sliding graph windows improved Android visualization stability

FULLY ASSEMBLED SPINNING-TOP



Full project & source code available at:

https://github.com/liyakesselman/Spinning_Top_Real_Time_Platform